Problem set 1: Simulating Customer Clicks Using a Bernoulli Random Variable

Problem statement

An organization sends promotional emails. Historical data indicate that a customer clicks the link with probability $p = 0.30$.

Define:

$$X = \begin{cases} 1, & \text{customer clicks} \\ 0, & \text{customer does not click} \end{cases}$$

Thus,

$$X \sim \text{Bernoulli}(0.30)$$

tasks

Step 1: Identify the random variable

Answer the following:

1. What does $X = 1$represent?
2. What does $X = 0$represent?
3. What are the possible values of X?
4. Is Xa discrete or continuous random variable?
5. Which probability distribution is suitable for X?

Step 2: Write the theoretical PMF

Step 3: Generate outcomes for 20 customers. Hint: np.random.binomial(n=1, …) generates Bernoulli observations because only one trial is performed for each customer.

Use code: samples = np.random.binomial(n=1, p=p, size=sample_size)

Answer the following:

1. How many observations were generated?
2. What does each zero represent?
3. What does each one represent?
4. Are your generated observations the same as the theoretical probabilities?

Step 4: Count clicks and non-clicks

Complete the code

number_of_clicks = np.sum(samples == _____)

number_of_non_clicks = np.sum(samples == _____)

print("Number of clicks:", number_of_clicks)

print("Number of non-clicks:", number_of_non_clicks)

Answer:

- How many customers clicked?
- How many customers did not click?
- Do the two counts add up to 20?

Step 5: Calculate the empirical PMF

Step 6: Compare empirical and theoretical PMFs

Answer the following:

1. Is the empirical probability of a click exactly 0.30?
2. Which outcome has the larger probability?
3. Why are the empirical and theoretical probabilities different?

Step 7: Plot empirical and theoretical PMFs

Step 8: Calculate the theoretical expectation

Step 9: Calculate the empirical expectation

Step 10: Calculate the theoretical variance

Step 11: Calculate the empirical variance

Step 12: Prepare the final result table

Measure	Theoretical value	Empirical value	Absolute difference
$P(X = 0)$	_____	_____	_____

Measure	Theoretical value	Empirical value	Absolute difference
$P(X = 1)$	_____	_____	_____
Expectation	_____	_____	_____
Variance	_____	_____	_____

Step13: Check Effect of sample size, Repeat the experiment n=10, 100, 1,000, 10,000

Sample size	Empirical $P(X = 0)$	Empirical $P(X = 1)$	Empirical mean	Empirical variance
10				
100				
1,000				
10,000				

Problem Set 2: ML Use Case: Number of Fraudulent Transactions Flagged

Problem statement: A fraud-detection model examines 50 transactions every hour. Assume that each transaction has a 4%probability of being flagged as potentially fraudulent.

Let:

$$X = \text{number of transactions flagged in one hour}$$

Initially assume that individual flagging decisions are independent.

$$X \sim \text{Binomial}(n, = 50p = 0.04)$$

The fraud investigation team can examine at most four flagged transactions per hour.

Find:

1: Simulate the operation of the fraud-detection system for 20,000 independent one-hour periods.

During every simulated hour:

- exactly 50 transactions are examined;

- each transaction has a flagging probability of 0.04;
- X records the total number of flagged transactions.

Hint: use np.random.binomial function